



Nirbhay Singh

AWS | EKS | Terraform | CI/CD
CloudFormation | SageMaker | Cost Optimisation

CONTACT

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CORE SKILLS

AWS COMPUTE

EC2, EKS, ECS, Lambda, Batch, Spot Instances, ASG, Fargate

AWS STORAGE & DB

S3, RDS, Redshift, DynamoDB, EBS, EFS, Glacier, ElastiCache

AWS NETWORKING

VPC, Route53, CloudFront, ALB/NLB, Transit GW, Direct Connect, PrivateLink

AWS AI/ML & DATA

SageMaker, EMR, Glue, Athena, Kinesis, Redshift, Lake Formation

AWS SECURITY

IAM, KMS, CloudTrail, GuardDuty, Security Hub, Config, Organizations, SCPs

IAC & CI/CD

Terraform, CloudFormation, CDK, GitHub Actions, Jenkins, CodePipeline, ArgoCD

CONTAINERS

EKS, ECR, ECS Fargate, Docker, Helm

OBSERVABILITY

CloudWatch, X-Ray, Prometheus, Grafana, ELK

LANGUAGES

Python, Bash, YAML, JSON, HCL, PowerShell

EDUCATION

MTech (Master of Technology)

Birla Institute of Technology

LANGUAGES

English	Professional
Hindi	Native

Nirbhay Singh

AWS DevOps Architect | AWS Solutions Architect - Professional

11+

Years Experience

\$45K+

AWS Cost Savings

30x

CI/CD Speedup

11,000+

Servers Migrated

14+

Certifications

PROFESSIONAL SUMMARY

AWS Solutions Architect Professional with 11+ years designing and operating production AWS infrastructure at enterprise scale -- EKS, EC2, RDS, S3, VPC, Batch, SageMaker, Lambda, CloudFormation. Cut AWS costs \$45K+, took a CI/CD pipeline 30x faster, orchestrated 1,000+ containers on EKS, and migrated 11,000+ servers into AWS. Now leads AWS architecture strategy at Bosch Europe, where FinOps governance has returned EUR 4M+ in documented savings.

KEY ACHIEVEMENTS

- Cut AWS costs \$45,000 at Deloitte through Spot Instance strategy, Reserved Instances, right-sizing, and pipeline optimisation
- Orchestrated 1,000+ containers on AWS Batch, Spot Instances, and EKS for distributed analytics and ML workloads
- Migrated 11,000+ on-premise servers into AWS with zero-downtime cutover and automated rollback
- Designed fault-tolerant, multi-AZ, cross-region AWS architectures holding 99.99% uptime SLAs in production
- Open-sourced InfraCents, posting real-time AWS cost estimates on every Terraform pull request

PROFESSIONAL EXPERIENCE

Cloud & AI Architect (AWS Lead)

Dec 2022 - Present

Robert Bosch (Bosch Polska) Warsaw, Poland

- Defined AWS architecture standards, landing zone design, and multi-account governance (Organizations, SCPs) for Bosch Poland's AWS partnership
- Designed AWS-native AI/ML infrastructure: SageMaker pipelines, EKS model serving, S3 data lake, and Glue ETL for enterprise workloads
- Built the FinOps governance model across 50+ AWS accounts behind EUR 4M+ in savings: Savings Plans, right-sizing, automated cleanup
- Made AWS Well-Architected reviews standard practice, mentoring architects on security, reliability, and cost optimisation

Senior DevOps Consultant

Jun 2021 - Nov 2022

McAfee

- Codified 300+ AWS resources in Terraform, cutting deployment time 50%+ with near-zero configuration drift
- Ran production EKS and ECS clusters with autoscaling, Spot integration, and GitOps deployments via ArgoCD
- Built AWS observability on CloudWatch, Prometheus, and Grafana for real-time infrastructure and application monitoring
- Enforced AWS security best practices: IAM least-privilege, KMS encryption, VPC flow logs, automated compliance scanning

Senior DevOps Engineer

Dec 2018 - Jun 2021

Deloitte

- Cut AWS costs \$45,000 through Spot Instance fleet management, Reserved Instances, Savings Plans, and right-sizing
- Redesigned CodePipeline around parallel stages and artefact caching, taking CI/CD 30x faster
- Orchestrated 1,000+ containers on AWS Batch, Spot Instances, and EKS for distributed data processing and ML training
- Built data and AI platforms on EMR, Redshift, and SageMaker for Fortune 500-scale analytics and model training

Senior Cloud Infrastructure Engineer

Nov 2015 - Dec 2018

PTC

- Designed fault-tolerant AWS architectures (EC2, RDS Multi-AZ, S3, VPC) for production SaaS workloads
- Led migration of 11,000+ on-premise servers into AWS with zero-downtime cutover and automated rollback
- Built disaster recovery with cross-region replication, automated failover, and validated RTO/RPO testing
- Designed AWS networking (VPC, subnets, security groups, NACLs, Transit Gateway) for multi-tier applications

System Engineer

Mar 2013 - Jul 2014

IBM

- Handled P1/P2 incident response and root cause analysis for enterprise production environments

OPEN-SOURCE PROJECTS

InfraCents -- AWS Cost Estimates on Every Terraform PR

>> github.com/nirbhays/infracents

Posts real-time AWS cost estimates on Terraform pull requests by parsing diffs against the live AWS Pricing API -- zero config, native PR integration.

STACK: Python, Next.js, Terraform, GitHub App, AWS Pricing API

ShieldIaC -- AI-Powered IaC Security Scanner

>> github.com/nirbhays/shieldiac

Scans Terraform and CloudFormation for AWS security misconfigurations pre-production against 100+ rules across 9 compliance frameworks (CIS AWS Benchmark, SOC2, HIPAA, PCI-DSS), with GPT-4.1 fix suggestions.

STACK: Python, Terraform, CloudFormation, GPT-4.1, GitHub Actions

TokenMeter -- LLM Cost Intelligence

>> github.com/nirbhays/tokenmeter

Tracks token cost and latency across LLM providers and routes by task complexity, cutting AI spend up to 60% with <5ms overhead.

STACK: Python, OpenAI, Anthropic, Google AI, Smart Routing, PyPI

CERTIFICATIONS & CREDENTIALS

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|---|---|
| ■ AWS Solutions Architect - Professional (Feb 2020) | ■ AWS Solutions Architect - Associate (Jan 2021) |
| ■ CKAD - Certified Kubernetes Application Developer (Valid Sep 2026) | ■ HashiCorp Terraform Associate |
| ■ Google Professional Cloud Architect (Valid Jun 2026) | ■ Google Professional Cloud DevOps Engineer (Valid Dec 2027) |
| ■ Google Professional ML Engineer (Valid Dec 2027) | ■ Google Generative AI Leader (Valid Aug 2028) |
| ■ Google Associate Cloud Engineer (Aug 2023) | ■ MCSA - Microsoft Certified Solutions Associate (Apr 2015) |